

Course Project Proposal

Project 1: Foundation Model for Speed Test Learning Tasks

Proposal Submission

1. Project Selection

Selected Project. Our group has selected Project 1: Foundation Model for Speed Test Learning Tasks. The goal is to study whether a pretrained foundation model over network measurement traces can improve early termination of Internet speed tests and support broader performance prediction tasks.

2. Proposed Understanding and Approach

Problem Framing. Traditional speed tests often run for a fixed duration even when enough evidence may already be available to estimate the final outcome. We plan to model each speed test as a multivariate network trace and learn from partial prefixes of that trace. The central question is whether a pretrained model can decide when a test can stop early while still predicting what the final speed would have been if the test had continued normally.

Primary Dataset. Our main dataset will be M-Lab NDT/ndt7, which is the same dataset family used in the reference paper. This makes it the most suitable benchmark for direct baseline comparison and for reproducing the overall problem setting.

Optional Supporting Datasets. If time permits, we may additionally explore FCC MBA, RIPE Atlas, and CAIDA as optional supporting datasets for transfer evaluation, robustness analysis, or broader validation of learned representations. These datasets will not be the core basis of the project.

Input Representation. We have not finalized the Transformer input format yet. Our current plan is to represent each speed test as a sequence of fixed time windows or patches. Candidate inputs include instantaneous throughput, cumulative throughput, elapsed time, RTT-related signals, retransmissions or loss indicators, congestion-related statistics, and other derived temporal trends depending on feature availability in the dataset.

Modeling Plan. Instead of using a general LLM, we will build a pretrained foundation model for multivariate network traces. The initial model will likely be Transformer-based and pretrained using self-supervised objectives such as masked-window reconstruction, next-window forecasting, or prefix-to-future prediction. After pretraining, the model will be fine-tuned on multiple downstream tasks related to speed testing.

Downstream Tasks. The main downstream task will be early termination: deciding whether a speed test can stop safely at a given point. We will also study final throughput prediction, speed-grade classification, and, if feasible, related tasks such as identifying unstable traces or cases where a test should not be terminated early.

Architecture Exploration. A major part of this project will be design exploration. We plan to experiment with channel-aware embeddings, relative time encodings, patch-based tokens, shared encoders with multi-task heads, and possibly confidence or uncertainty-aware outputs. Because the exact architecture is still open, our early experiments will help determine the final design choices.

Evaluation Plan. Baseline comparison will be done against the paper provided for this project. We will compare pretrained and non-pretrained variants and evaluate both prediction quality and measurement savings. Planned metrics include relative error and data reduction for early stopping, MAE or RMSE for final throughput prediction, and accuracy or F1-score for speed-grade classification. We will also analyze performance across different speed ranges and network conditions.

Broader Implications. Beyond one stopping decision, we want the learned model to provide a reusable representation for other network measurement and performance prediction problems. In particular, the same pretrained model may help with predicting final speed from early observations, grading connection quality, and identifying when longer testing is necessary.

3. Project Plan and Timeline

Planning Note. We intentionally compress the initial setup phase so that more time can be spent on model design, training, testing, and architectural experimentation, which are expected to be the main research contributions of the project.

Week	Planned Milestone
Week 1	Literature review, dataset access, and preprocessing plan.
Week 2	Trace extraction, prefix construction, and baseline implementation.
Week 3	Baseline reproduction and Transformer input representation design.
Week 4	Self-supervised pretraining setup and first model prototype.
Week 5	Architecture experiments and representation refinement.
Week 6	Fine-tuning for early stopping, throughput prediction, and speed-grade tasks.
Week 7	Training, testing, ablation studies, and robustness analysis.
Week 8	Final evaluation, result consolidation, report writing, and presentation preparation.

References

[1] H. Manda et al., "TURBOTEST: Learning When Less is Enough through Early Termination of Internet Speed Tests," arXiv:2510.21141, 2025.

[2] Measurement Lab, "NDT / ndt7 documentation" and public data documentation for the NDT dataset.

[3] FCC Measuring Broadband America raw data releases (optional supporting dataset).

[4] RIPE Atlas documentation and public measurement data resources (optional supporting dataset).

[5] CAIDA passive Internet trace datasets (optional supporting dataset).